➢ Within 100 years, **climate change increases the amount of carbon stored in aboveground biomass**, refuting our hypothesis that climate change will result in decreased carbon storage in the forests of Petersham, Massachusetts.

Climate Change, Tree Harvest and Carbon Storage in New England with LANDIS-II

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Introduction: Forests in New England

Forests cover about **33%** of land on Earth. Services forests provide:

- Water purification
- Habitat structure
- Biodiversity
- Carbon sequestration

(Hou et al, 2023; Bonan, 2008)

Forests store nearly **50%** of all terrestrial carbon (Bonan, 2008)

Following disturbance events like natural disasters and timber harvesting, temperate forests have shown faster biomass accumulation (Anderson-Teixeria et al., 2013)

Biomass accumulation means an increase in forest carbon storage (Vashum & Jayakumar, 2012)

80% of forested land in New England is private land, meaning it is logged privately (Thompson et al, 2017).

In the US, private land is twice as likely as public land to be harvested each year.

Privately logged land is more difficult to regulate than public land harvesting.

This can create debate over management, land use, and policy (Bellassen & Luyssaert, 2014).

Climate Change

- Over the last two decades, global temperatures have risen 1.53 C more than the industrial levels (1850-1900). (IPCC, 2019)
- ~46% rise in carbon dioxide levels, and greenhouse gases (IPCC, 2019)
- Climate change leads to longer growing seasons, warmer temperatures, and increased moisture, which can all **promote tree growth**, meaning **more carbon storage in the short term**.

Question: How does tree harvest affect carbon storage in the context of climate change, and how can LANDIS-II be used to investigate this relationship?

Hypothesis: By using the simulation model, LANDIS-II, we will show that climate change decreases the amount of carbon that is stored in aboveground biomass and amplifies the effects of tree harvest.

Why Environmental Modeling?

- Allows people to visualize and prepare for future scenarios
- Safe, non-destructive method to simulate harvest effects without needing to conducting the harvest
- Modeling and computing potential is high, which allows scientists to simulate very complex interactions

Citations and Modeling Files

Please see this GitHub link for information on literature and simulation input files:
https://github.com/maura-collins/SCE_landis

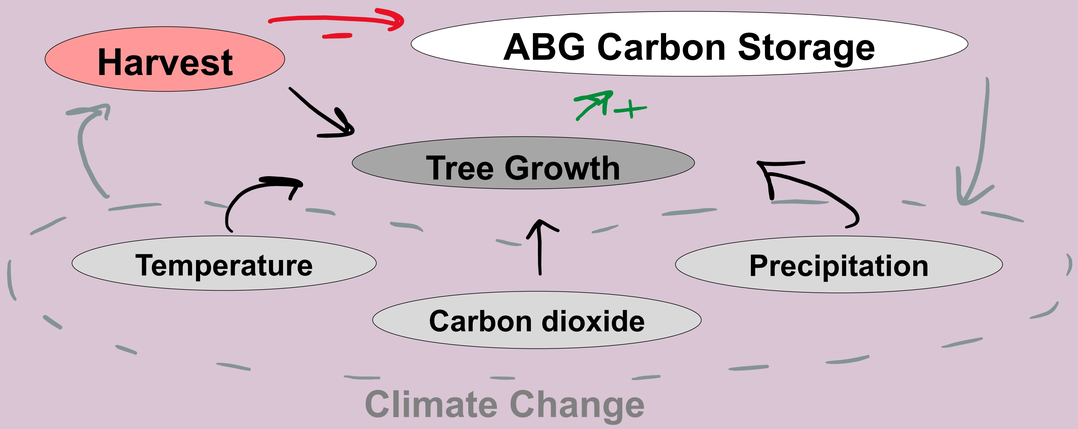


Figure 1. This diagram depicts the effect of harvest on tree growth and aboveground (ABG) carbon storage. Represented in black are the direct effects of climate change on tree growth, and in grey are the indirect interactions of climate change, harvest, and carbon. In this study, climate change will be represented by temperature, carbon dioxide and precipitation.

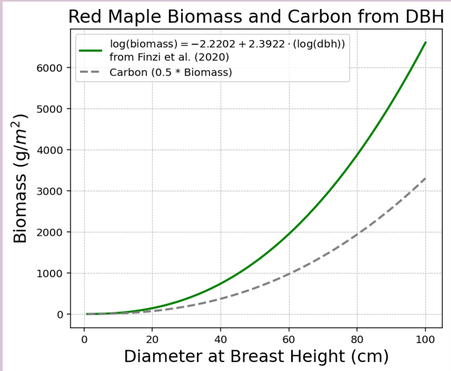


Figure 2. As the diameter of the tree increases, as does the estimated biomass, and also the estimated carbon.

Eq 1: Tree Growth and Carbon

- Tree growth, as biomass (g/m²), can be modeled with established relationships between other physical characteristic, or *allometric equations*
- Diameter at breast height 'dbh' (cm) is the diameter of a tree 1.3m above the ground
- The relationship between biomass and dbh, for the species red maple (*acer rubrum)* can be expressed as (Finzi et al, 2020):

$$\textbf{EQ 1: } \text{biomass} = 0.1086(dbh)^{2.3922}$$

- Carbon accounts for approximately 45-50% of overall biomass, so half the biomass estimation is the approximate amount of carbon contained in a gram per meter squared of biomass.

Eq 2: Growth and Harvest

- Harvest is traditionally represented with a sigmoid 'S' curve, where the top is "harvested" and the bottom is "not harvested"
- Nuanced models show that the likelihood an individual tree is harvested is dependent on the (i) **species,** (ii) the **tree size** (dbh) and (iii) the **percentage of the forest that has already been harvested**. Canham et al. (2013) models (i/iii) as:

$$\textbf{EQ 2: } F(R) = 1 - \gamma_s \, exp(-\beta_s R^{\alpha_s})$$

R represents the percentage of forest already harvested, $\gamma_s, \beta_s, \propto_s$ (scaling, rate, curvature) represent species specific parameters, and overall *F(R)* is the probability that any tree is selected to harvest.

- Equation 2 represents half of the full regression model, which factors in biomass to estimate likelihood of a tree being logged. **As biomass increases, the likelihood of being harvested occurs**

LANDIS Modeling Discussion

Result: Within 100 years, **climate change increases carbon stored in aboveground biomass for forests in the study area**, refuting our hypothesis.

- LANDIS simulated several species level differences in tolerance for climate change and harvest disruption, seen in literature (Thompson et al, 2013; Wang et al., 2019).
- Indicates that 40% clear cut harvest is intensive, and species-level carbon forestry is necessary even for species well within their range (Thompson et al., 2013)

Limitations: 30m resolution a highly generalized 90m² stand, Longer simulation period could yield more understanding of long-term dynamics.

Future Directions: Measuring Harvest Success with Basal Area Increment; Including replanting; introducing more complex harvest regimes to better mimic current practices.

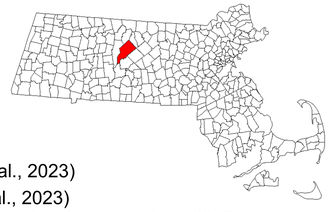
Methods: Simulating Harvest, Climate Change and Carbon Storage

LANDIS-II: Forest Landscape Model

Study Area

- Petersham, Massachusetts, USA; 14,000 hectares, 90% forested
- 5 month growing season, 105 cm annual precipitation , mean annual temp is 8.5C
- Dominant tree species include American Hemlock, Red Maple, Eastern White Pine and Red Oak
- Timber harvest is managed by Harvard Forest and private woodlots

Figure 3. Location of the study area is within Massachusetts. Petersham is highlighted in dark red and located at (42.49 N, -72.19 W)



LANDIS-II Extensions

- PnET Succession (Gustafson et al., 2023)
- Biomass Harvest (Gustafson et al., 2023)
  - 1 management area, 40% clearcut (Thompson et al., 2017)
  - 30 x 30-meter pixel resolution, each pixel is an individual stand

Simulation Design

- Run simulations from 2010-2110.
- 2 Controls: No harvest (Static Climate, RCP 8.5 - climate change)
- Treatment: Harvest prescription with Static and RCP 8.5
- Compare forest aboveground biomass totals at the 100-year mark

Simulation Results

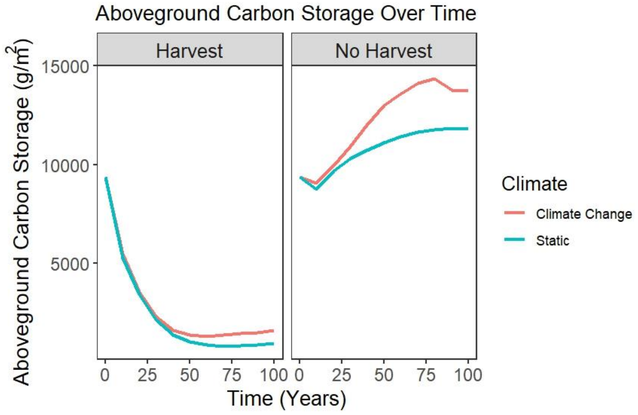


Figure 4. Harvest: Climate change yields 63% higher carbon storage than the static scenario. No Harvest: Climate change yields 16% higher carbon storage.

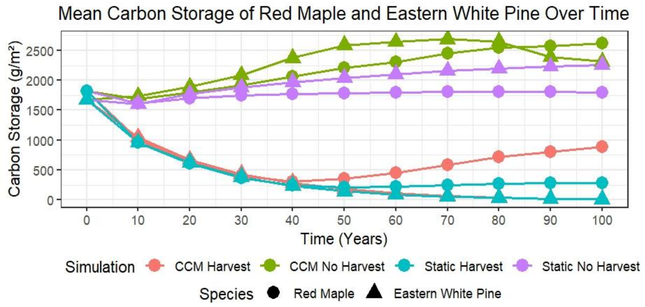


Figure 5. Demonstrating species level differences: across all four simulations, Eastern White Pine only acquired higher carbon storage than Red maple in the simulation with static climate, and no harvest.

Acknowledgements

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